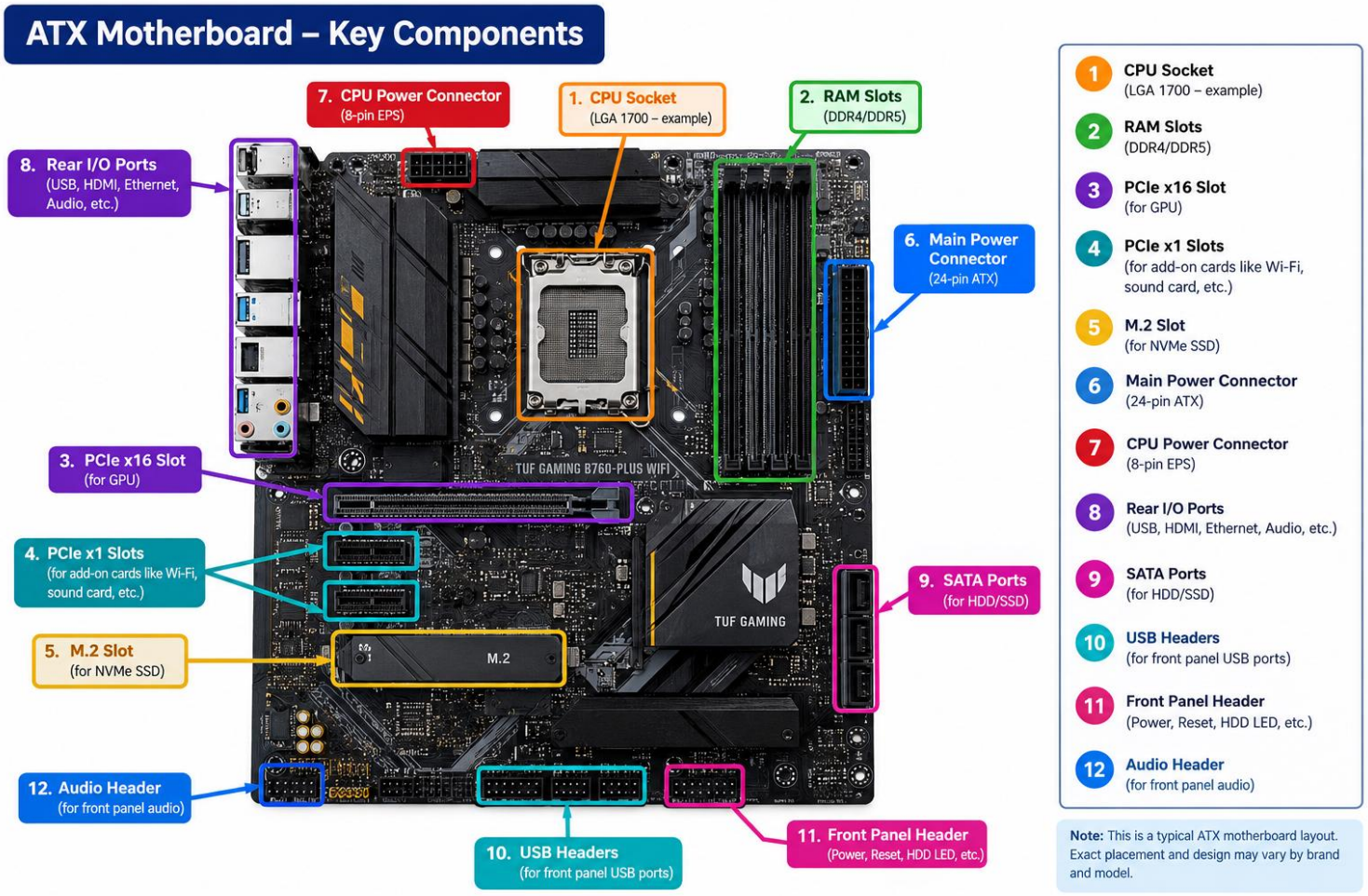
Day-2

Motherboard and CPU Theory

- Types of motherboard and CPU socket compatibility.
- Functions of the CPU and chipset .
- Identify the motherboard slots and ports.

Q 1 : Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports(such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket,power connector, etc.).

Ans- An ATX motherboard is a standard-sized main circuit board inside a desktop computer that connects components such as the CPU, RAM, GPU, storage drives, and power supply so they can work together. The diagram of an ATX motherboard as below.



Q 2 : Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

Ans -

CPU Socket Type	Compatible CPU Model	Details
LGA 1151	Intel Core i7-9700K	Has 8 cores and is good for gaming and multitasking.
AM4	AMD Ryzen 5 5600X	Has 6 cores and provides good gaming and everyday performance.
LGA 1200	Intel Core i7-10700K	Has 8 cores and is suitable for gaming and video editing.



1. LGA 1151 (Intel Core i7-9700K)

An Intel desktop processor designed for demanding programs and games.



2. AM4 (AMD Ryzen 5 5600X)

An AMD processor that balances gaming performance and everyday tasks.



3. LGA 1200 (Intel Core i7-10700K)

An Intel processor suitable for gaming, multitasking, and editing Videos.

Q 3 : Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

Ans-

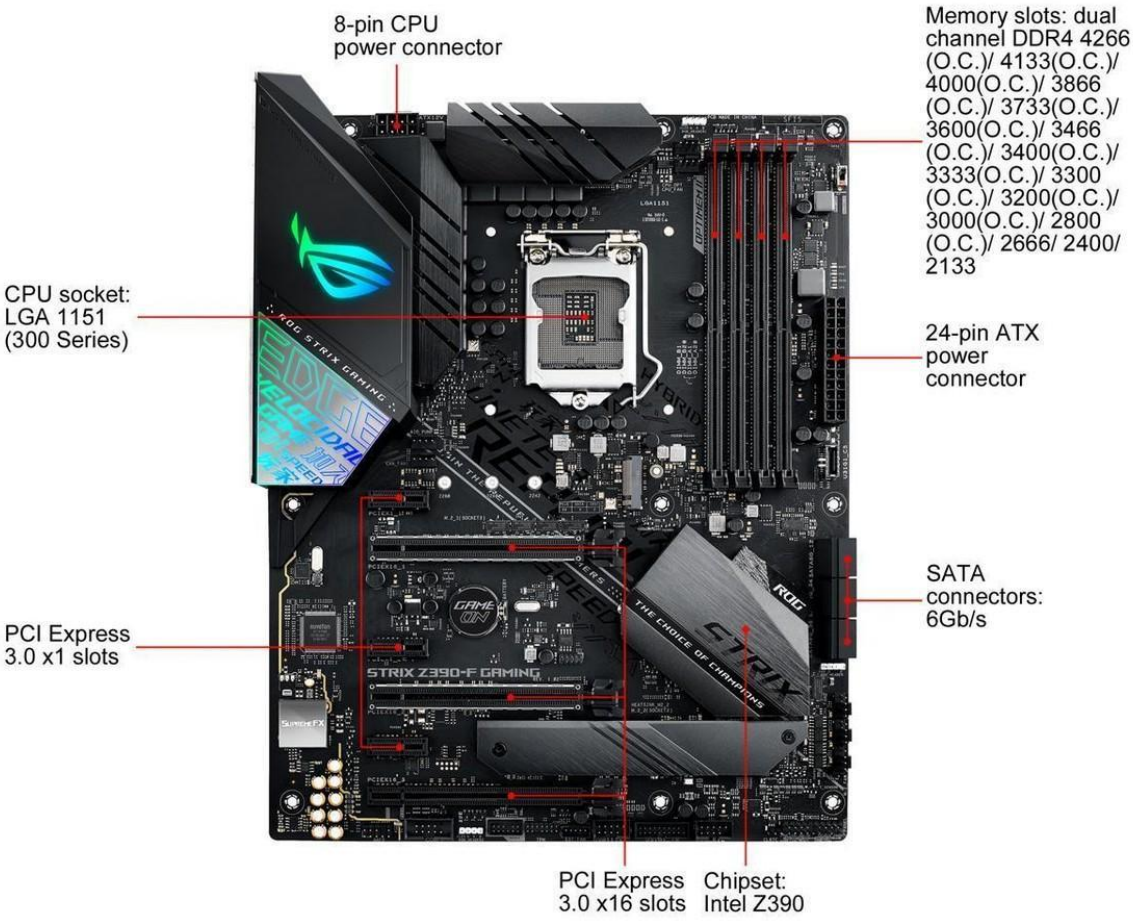
1. Primary functions of the CPU :-
The CPU (Central Processing Unit) is the brain of the computer. It processes instructions, performs calculations, and tells other components what to do. When you open Spotify, the CPU processes the app's instructions and helps it run smoothly.

2. Primary functions of the chipset :-
The chipset helps manage communication between the CPU and other parts of the motherboard, such as USB ports, storage devices, and other connected hardware. When you open Spotify, it helps coordinate communication with devices and connections the app needs.

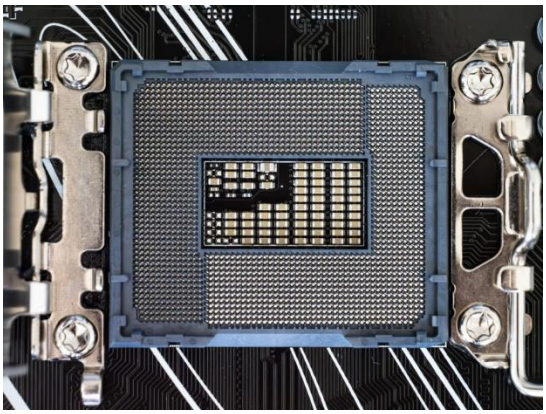
3. How they work together when opening Spotify :-
(1) You click the Spotify icon.
(2) The CPU processes the instructions needed to start Spotify.
(3) The chipset helps manage communication between the CPU, storage, and other connected devices.
(4) The app loads, and you can choose a song to play.

Q 4 : Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.

Ans-

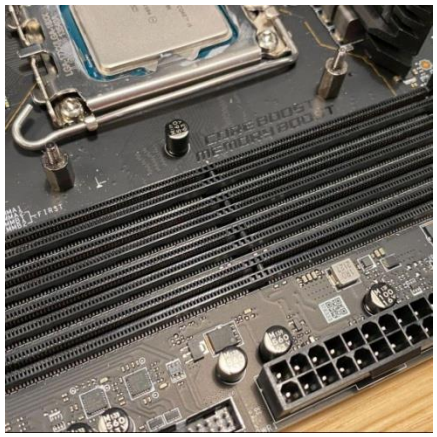


Identify these motherboard parts :



1. CPU socket :-

Holds the processor, which carries out computer instructions.



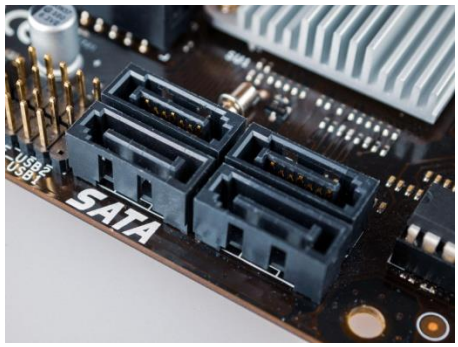
2. RAM slots :-

Hold memory sticks that help the computer run programs.



3. PCI / PCIe expansion slots :-

Allow extra cards, such as graphics or network cards, to be installed.



4. SATA ports :-

Connect storage drives, such as HDDs and SATA SSDs.



5. Main power connector :-

Supplies electrical power from the power supply to the motherboard.